AALIYA KHAN

II Year UG Student, Electronics and Instrumentation Engineering, SGSITS, Indore, M.P., India

aaliyakhanonly25@gmail.com | +91 7724085878 | linkedin.com/in/aaliya-khan2425

About Me

Second-year Electronics and Instrumentation student with a strong and growing interest in quantum computing, post-quantum cryptography, and AI/ML-driven intelligent systems. Recently completed a research internship at iQCMAT (Centre of Excellence in Quantum Computing, Materials, Algorithm and Technology), SGSITS Indore, working on post-quantum secure communication and quantum algorithm design, alongside academic and project experience applying machine learning to real-world, hardware-driven problems.

Education

SGSITS, Indore — *Aug 2025 – Sep 2029*
Bachelor of Technology in Electronics and Instrumentation Engineering

The Juniors' College, Indore — *Jun 2022 – Jun 2024*
Higher Secondary Education

Sanmati H.S. School, Indore — *Apr 2015 – Jun 2022*
Secondary Education

Skills

Programming & Tools	Python, HTML, CSS, React.js, Git & GitHub, MATLAB, Simulink, LaTeX
Libraries	NumPy, Pandas, Scikit-learn, Qiskit
Quantum Computing	Discrete-variable (gate-based) circuit design and simulation; quantum information fundamentals (superposition, entanglement, Bloch sphere); quantum gates (Pauli, Hadamard, CNOT, Toffoli); QKD protocols (BB84, decoy-state); post-quantum cryptography (lattice-, hash-, and code-based schemes; ML-KEM, ML-DSA)
Machine Learning	Supervised & Unsupervised Learning, Decision Trees, Random Forests, Clustering, Model Evaluation, Data Fusion
Hardware Interfacing	LabVIEW-based DAQ signal acquisition, analog photodetector integration, VISA session lifecycle
Operating Systems	Windows, Linux, macOS
Soft Skills	Analytical Thinking, Technical Communication, Team Collaboration, Adaptability, Ethical AI Awareness

Experience

Research Intern – Post-Quantum Cryptography & Quantum Algorithms — *Jun 2026 – Jul 2026*
iQCMAT, Department of Applied Physics and Optoelectronics, SGSITS Indore

- Worked across two tracks: Post-Quantum Cryptography & Secure Data Communication, and Quantum Solvers & Circuit Implementation
- Co-authored a research paper on a post-quantum secure optical data communication framework, covering system design, implementation, and experimental validation
- Implemented and studied quantum key distribution protocols (BB84, decoy-state) including QBER thresholding and their classical post-processing pipeline
- Built discrete-variable quantum circuits in Qiskit – Bell state generation, quantum half adder, and quantum teleportation (achieving fidelity = 1.0) – tested on IBM Quantum Composer
- Integrated NI DAQ hardware with analog photodetector signal chains in LabVIEW, covering TIA signal chains, DAQmx VI structure, and grounding modes for experimental QKD validation
- Reviewed NIST-standardized PQC schemes (ML-KEM, ML-DSA, SLH-DSA) and their role in defending against "harvest now, decrypt later" threats

Research Intern — *Sep 2025 – Present*
DRDO, in collaboration with MANIT Bhopal & IIT Indore

- Contributed to the development of a soft robotic hand gripper designed to handle 10–15 N load per finger for defence and healthcare applications
- Conducted an extensive literature review of 10+ research papers on soft robotics, gripping mechanisms, and sensor integration
- Explored and analyzed force, pressure, and tactile sensor technologies for optimizing grip performance
- Developed foundational skills in MATLAB and Simulink for modelling and simulation of robotic systems

Electric Vehicle & Web Dev Team Member – GS Motorsports *Sep 2025 – Mar 2026*
SGSITS Indore's official Formula Student racing team

- Contributed to the design and fabrication segment of the team's formula racing vehicle program

1st Place – PRARUPAM (Prototype Competition)
AAROHAN Techfest, SGSITS Indore

- Built PRAHAAR-AI, an AI-based defense intelligence system for threat detection, prediction, and battlefield visualization
- Applied data fusion and machine learning concepts to process multi-source sensor inputs

Hackathon Finalist – MANTHAN 1.0
Technosearch, MANIT

- Advanced to the final screening round with ROUTIQO, a unified delivery intelligence platform coordinating routes, real-time traffic, parking availability, and on-ground execution

Conferences & Workshops

- Attended a 5-day GIAN workshop at IIT Indore on *"Mechanics of Soft Active Materials and Higher Maths"*
- Participated in TiEcon MP, MP Startup Summit 2026 (Bhopal), and YEF Bharat to explore the industrial applications of AI/ML, sponsored by the Institute Innovation Council (IIC)

Projects

Post-Quantum Secure Optical Data Communication Framework
Research Paper, iQCMAT

- Designing, implementing, and experimentally validating an optical communication framework secured with post-quantum cryptographic primitives
- Combined QKD protocol simulation with classical PQC key-exchange schemes to build an end-to-end secure communication pipeline

PRAHAAR-AI – AI-Powered Battlefield Command & Control System *1st Place, PRARUPAM*
Aarohan Techfest

- Applied machine learning concepts and data fusion techniques for real-time threat analysis
- Built an interactive, map-based dashboard for real-time visualization; led frontend development and geospatial (map) integration
- Designed a secure and scalable system architecture
- **Tech:** Python, Scikit-learn, OpenCV, JavaScript, React.js, Mapbox API / Leaflet, HTML, CSS, NumPy, Pandas, Git